

Comparing Adverse Event Rates Across Treatment Groups in a Crohn’s Disease Trial

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Introduction

Crohn’s disease is a chronic inflammatory bowel condition characterized by periods of flare and remission. This report presents an exploratory analysis of a clinical trial evaluating three treatments—placebo, drug d1, and drug d2—with the number of adverse events (nrAdvE) as the primary response variable. The central question of interest is: does the number of adverse events differ significantly across treatment groups?

The data come from CrohnD, a dataset included in the R package robustbase, originating from the trial described in Cantoni and Ronchetti (2009) (Table 1). Each row represents a unique patient. Treatment groups are balanced, with 39 patients assigned to each of placebo, d1, and d2. However, the sample is heavily female (100 F, 17 M) and geographically skewed (78 from country c1, 39 from c2). These imbalances are not randomly distributed across treatment groups and warrant attention: if sex or country are associated with adverse event frequency, their unequal representation could confound treatment comparisons. The dataset originates from a single methodological illustration with no documented randomization scheme for these baseline characteristics, which limits the strength of any causal inference drawn from this analysis.

Table 1: Variable descriptions. Anonamized dataset from Genexion SA, Geneva courtesy of Dr. Enrico Chavez (Cantoni and Ronchetti, 2009).

Variable	Type	Description
ID	integer	Unique identifier for each patient
nrAdvE	integer	Number of adverse events experienced
BMI	double	Body Mass Index (kg/m^2)
height	integer	Height of the patient (cm)
country	factor	Country the patient is from (anonymized; c1, c2)
sex	factor	Sex of the patient (M or F)
age	integer	Age of the patient (years)
weight	integer	Weight of the patient (kg)
treat	factor	Treatment group (placebo, d1, d2)

Exploratory Data Analysis

Distributions of baseline characteristics (BMI, height, age, and weight) are shown in Figure 1. The sample skews toward middle-aged and older adults, with most patients between 40 and 65 years, consistent with the typical age of Crohn’s disease onset and progression. BMI is concentrated in the overweight range ($25\text{--}30 \text{ kg}/\text{m}^2$) with a right tail extending toward obesity, suggesting this characteristic may warrant consideration in modeling. Height and weight are roughly symmetric and do not show notable anomalies.

The response variable nrAdvE is a non-negative integer count ranging from 0 to 12 (Figure 2). Its distribution is strongly right-skewed, with 50 of 117 patients (43%) reporting zero adverse events and a long tail of patients reporting progressively higher counts. This shape is typical of clinical adverse event data and has immediate implications for statistical testing—the normality assumption required for ANOVA is unlikely to hold.

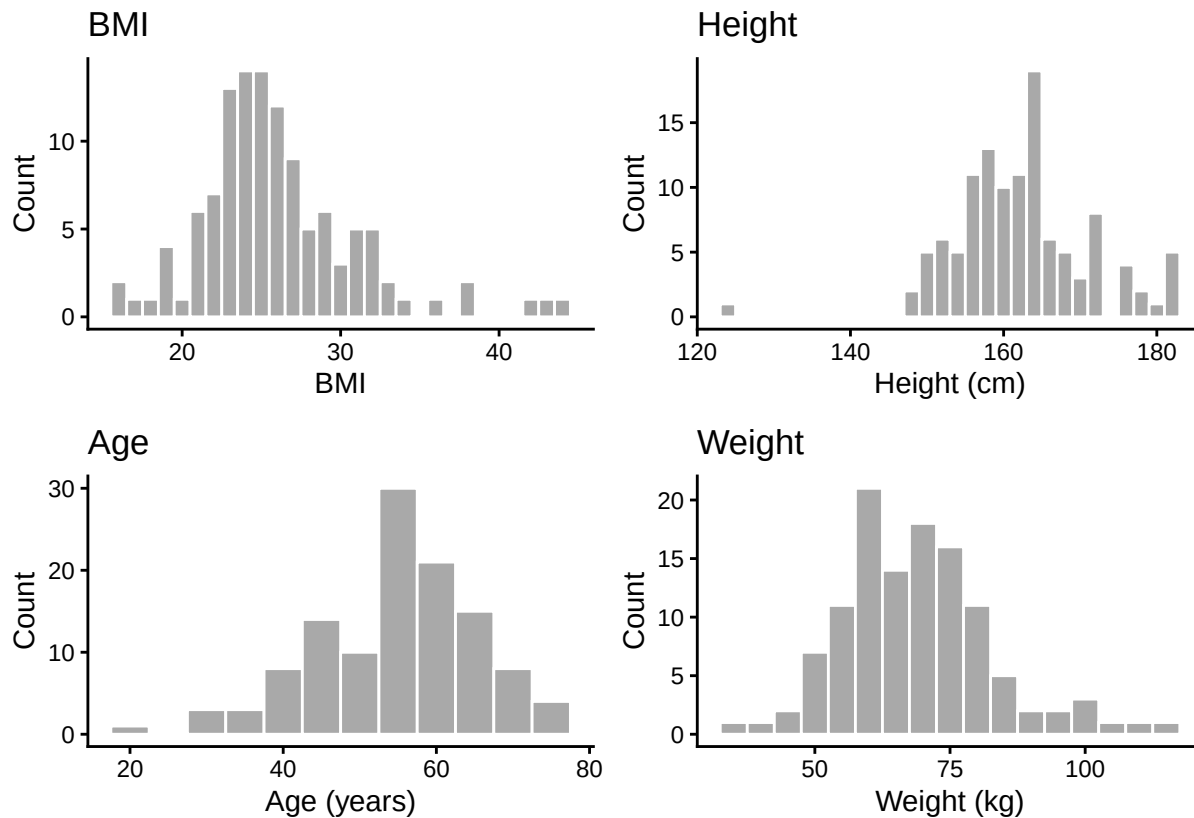


Figure 1: Distribution of baseline characteristics: BMI, height, age, and weight

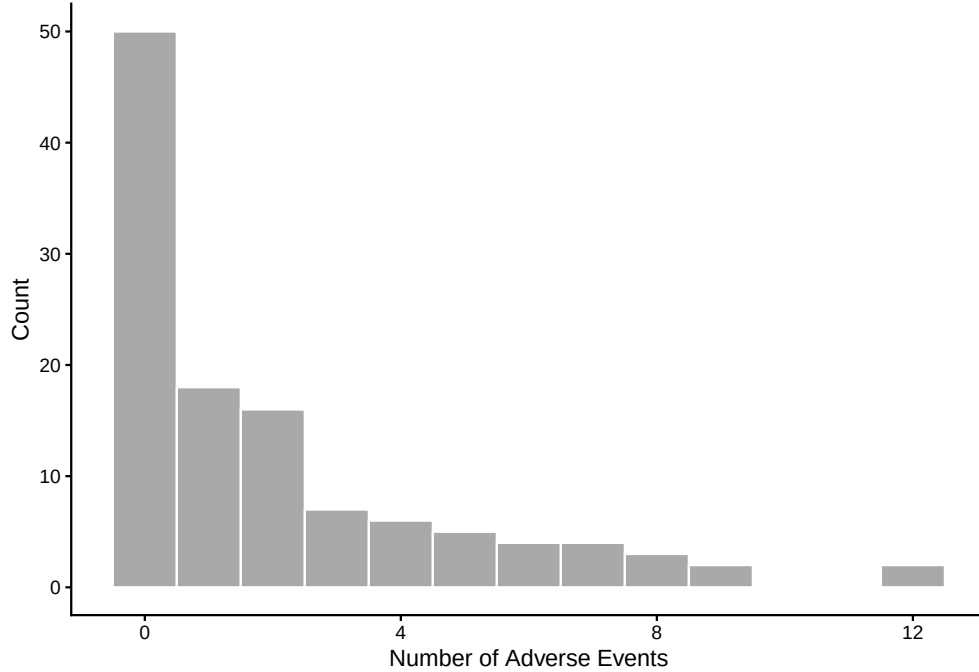


Figure 2: Distribution of adverse events (nrAdvE)

When faceted by treatment group (Figure 3), all three groups retain this right-skewed shape, but differ in concentration at zero. The d1 group shows the sharpest spike at zero relative to placebo, which shows greater spread across higher counts. The d2 group falls between the two. These visual differences motivate a formal comparison of nrAdvE across treatment groups.

Methods and Modeling

To assess whether the number of adverse events differs across treatment groups, a Kruskal-Wallis rank sum test was used. Prior to testing, the normality assumption required for a one-way ANOVA was evaluated using the Shapiro-Wilk test. The overall distribution of nrAdvE was found to be significantly non-normal ($W = 0.76$, $p\text{-value} < 0.001$), and this held within each treatment group individually (all $p\text{-values} < 0.001$). Given this violation, the non-parametric Kruskal-Wallis test was selected as the alternative to ANOVA. The null hypothesis is that the distribution of adverse events is identical across all three treatment groups. Observations are assumed to be independent, with each row representing a unique patient with no repeated measures. While Kruskal-Wallis is used here as an exploratory tool, the data source (Cantoni & Ronchetti, 2009) uses this dataset primarily as a methodological illustration and does not document a formal randomization scheme or sampling methodology, further supporting a descriptive rather than inferential interpretation. Given the count nature of nrAdvE, a negative binomial regression model would be better suited to formal inference and is recommended for future work.

Table 2: Kruskal-Wallis Test: Adverse Events by Treatment Group

	Statistic	df	p-value
Kruskal-Wallis chi-squared	2.399	2	0.301

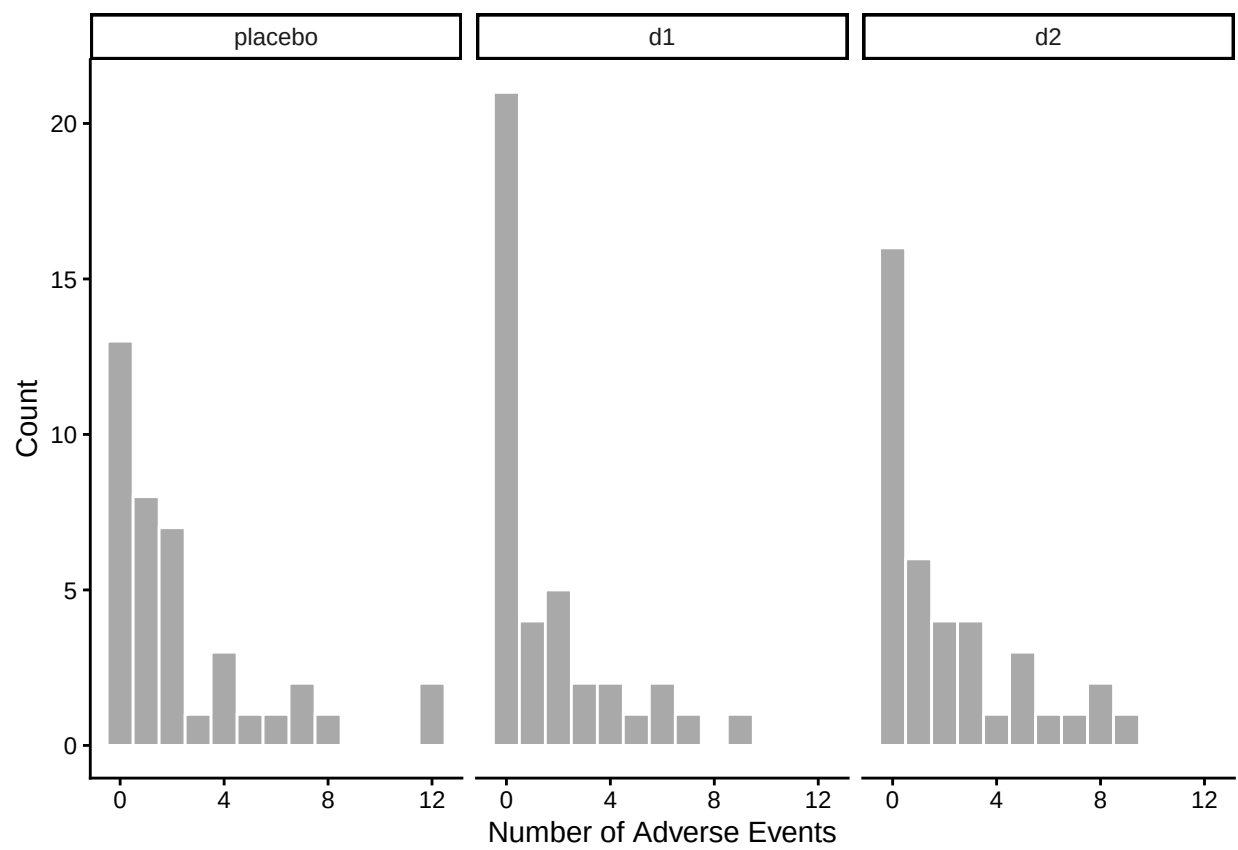


Figure 3: Distribution of adverse events by treatment group

Results

A Kruskal-Wallis rank sum test found no significant difference in the distribution of adverse events across treatment groups ($\chi^2 = 2.40$, $p = 0.301$). There is insufficient evidence to conclude that the number of adverse events differs between the placebo, d1, and d2 groups. This is consistent with the visual inspection of Figure 3, where all three groups show similar right-skewed distributions concentrated near zero, with overlapping spreads. Given the exploratory nature of this analysis and the inferential limitations noted in the Methods section, this null result should be interpreted cautiously rather than as definitive evidence of treatment equivalence.

Conclusions and Discussion

A Kruskal-Wallis test found no significant difference in adverse event counts across treatment groups ($\chi^2 = 2.40$, $p = 0.301$), suggesting that treatment assignment alone does not explain variation in nrAdvE in this sample. This analysis benefits from a balanced treatment design (39 patients per group) and the availability of multiple baseline characteristics that could support more nuanced modeling. However, several limitations temper these conclusions. The dataset is small ($n = 117$), heavily female (86%), and drawn from only two countries with no documented recruitment or randomization methodology, as noted from review of its primary source (Cantoni and Ronchetti, 2009). These factors limit both statistical power and generalizability. The Kruskal-Wallis test, while appropriate given the non-normality of nrAdvE, is a rank-based method that does not account for the count nature of the response variable or adjust for the observed imbalances in sex and country. A null result here should not be interpreted as evidence of treatment equivalence; with $n = 117$ and a right-skewed outcome, the test may lack sufficient power to detect a true effect. Future work should consider a negative binomial regression model, which would better accommodate overdispersed count data and allow adjustment for sex and country as potential confounders. Furthermore, the robust saddlepoint test proposed by Cantoni & Ronchetti (2009) for generalized linear models may warrant evaluation in future work, particularly with datasets where a formal randomization scheme is documented and the assumptions required for inference are met.

References

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